SELENIUM WEBDRIVER INSTALLATION

Selenium WebDriver installation process is completed in four basic steps:

1. Download and Install Java 8 or higher version.
2. Download and configure Eclipse
3. Download Selenium WebDriver Java Client
4. Configure Selenium WebDriver

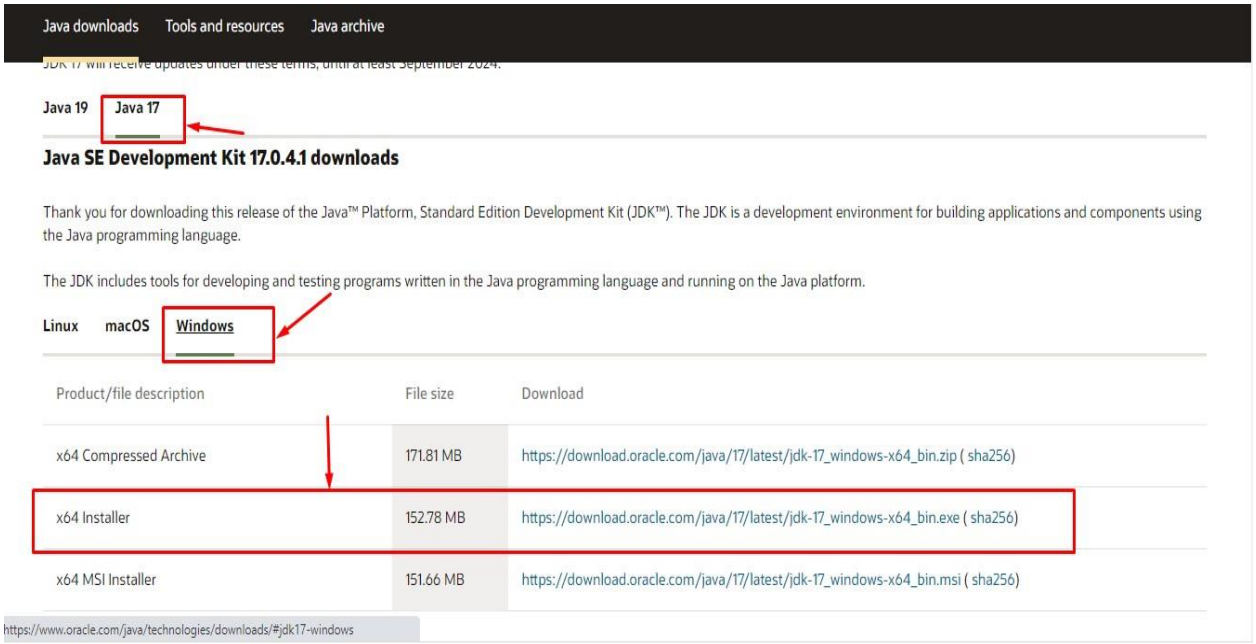
1. Download and Install Java:-

Download the latest version of Java Development Kit (JDK) from the link given below.

https://www.oracle.com/java/technologies/downloads/#jdk17-windows

1). **Select Java17----------> Select Windows-------->Select .exe file**

(as shown in the below image)

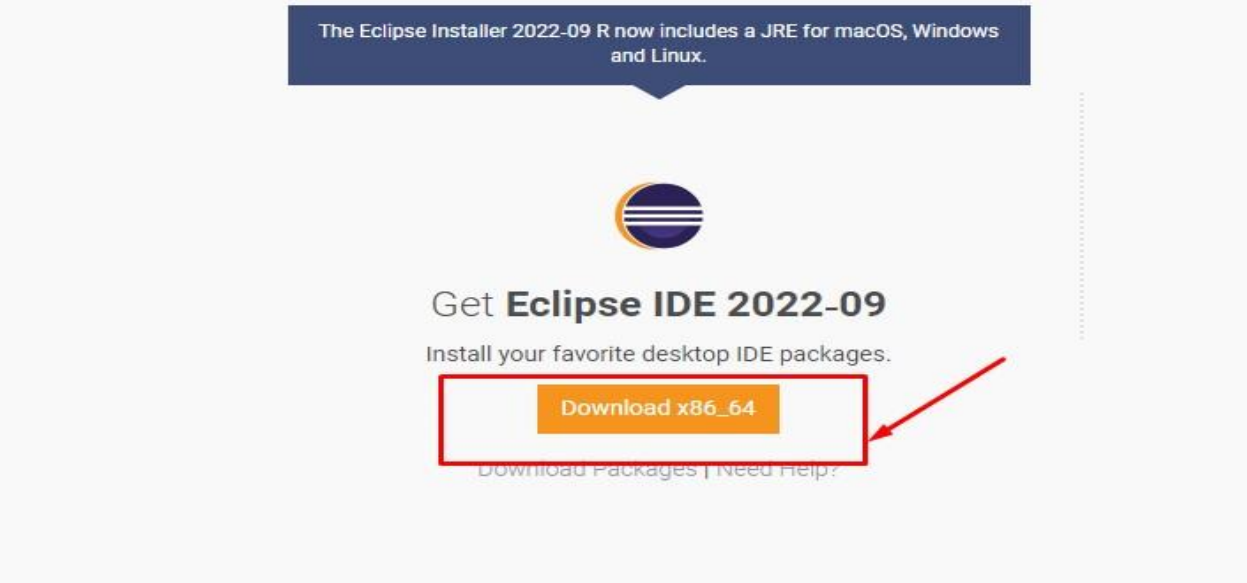


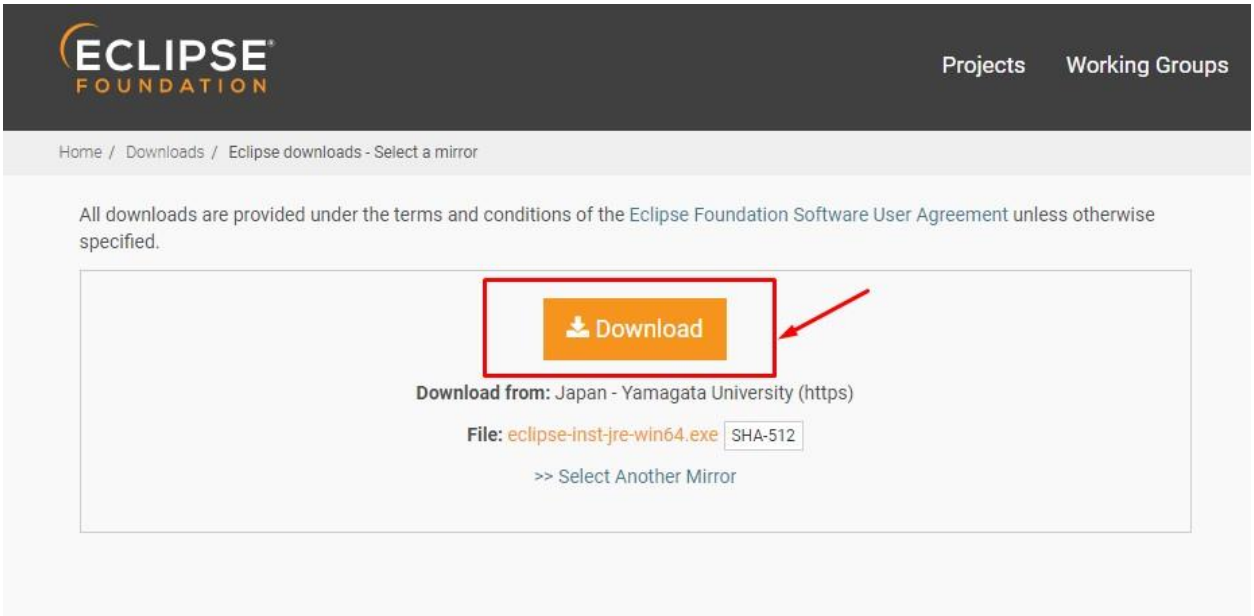
2). **Download the .exe file, then Double Click the file and Install.**

2. Download and Configure Eclipse IDE:-

1). **Open URL**: https://www.eclipse.org/downloads/ .

2). Select the Download Button and it will redirect to next page from there you can download the File. (as shown below)





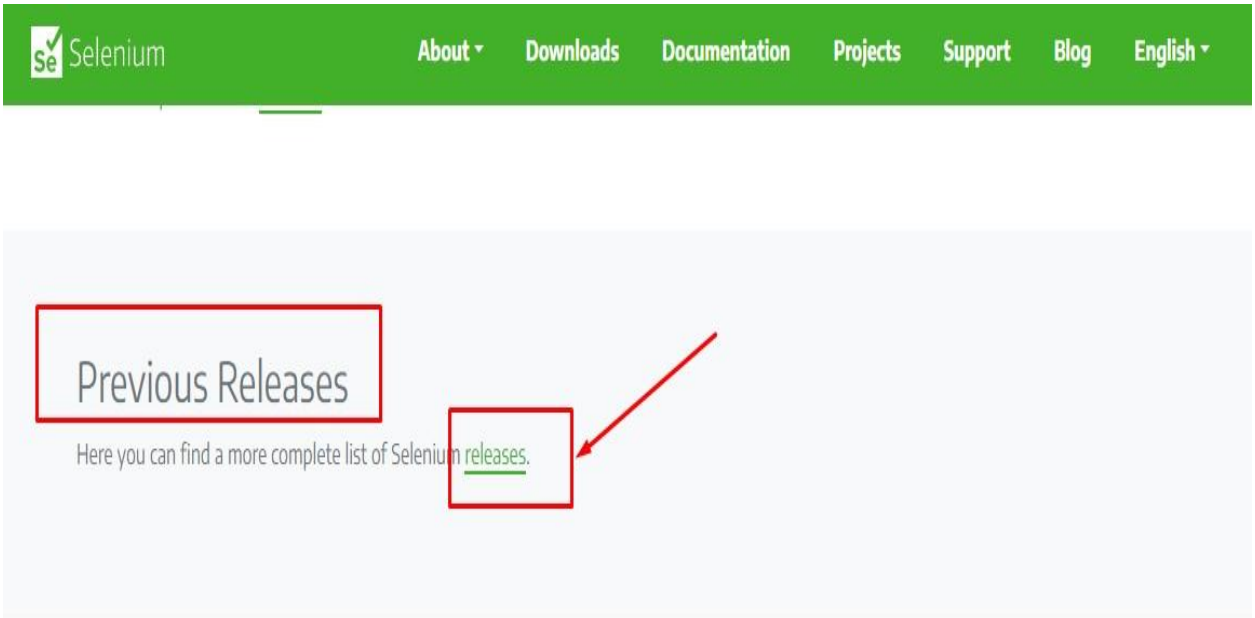
3). **Open the downloaded Eclipse application------>Select Eclipse for Java Developers------>Click Install------>Click Accept Now------>Click Launch.**

It will redirect you to a Welcome Window------>Select a **New Java Project**(Open a New Java Project)------>**Add Project Name**(First letter of Project should be in Caps).

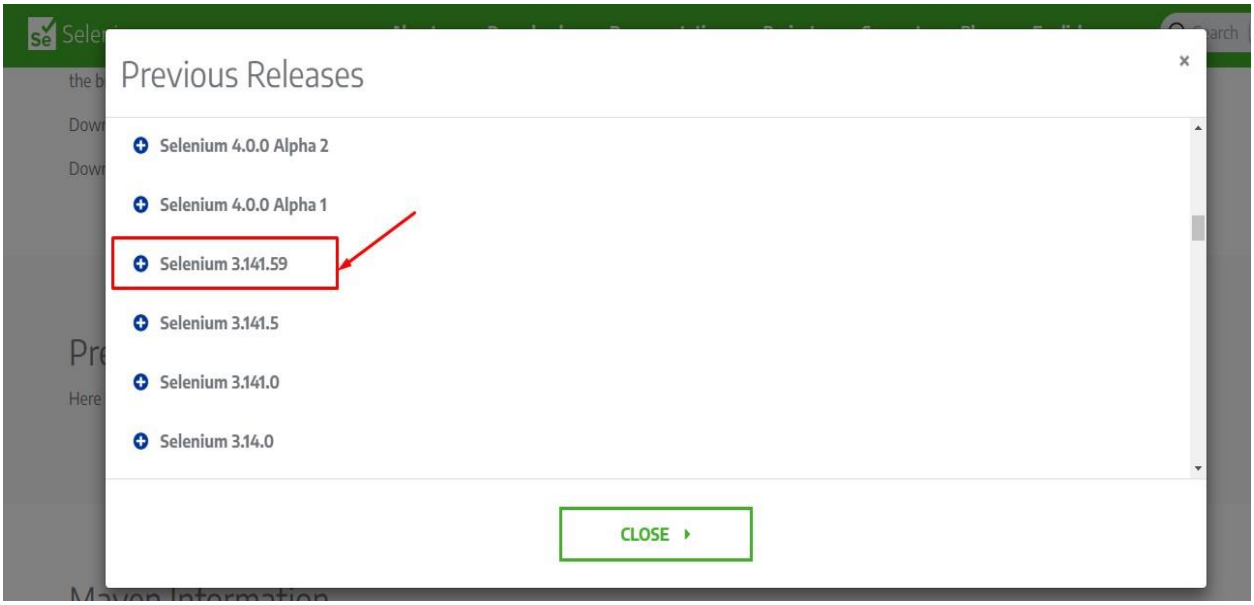
3. Download Selenium WebDriver Java Client

1). **OpenURL**: https://docs.seleniumhq.org/download/. It will redirect you to the "downloads page" of Selenium official website.

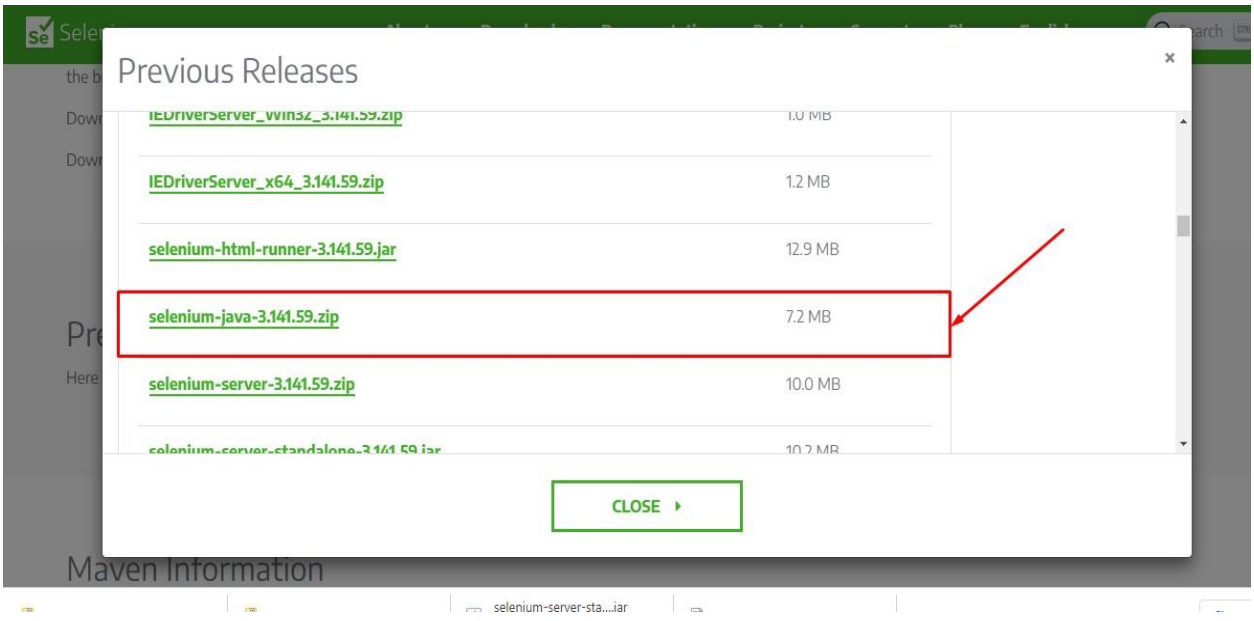
2). Scroll down through the web page and locate previous releases heading and click as shown below:-



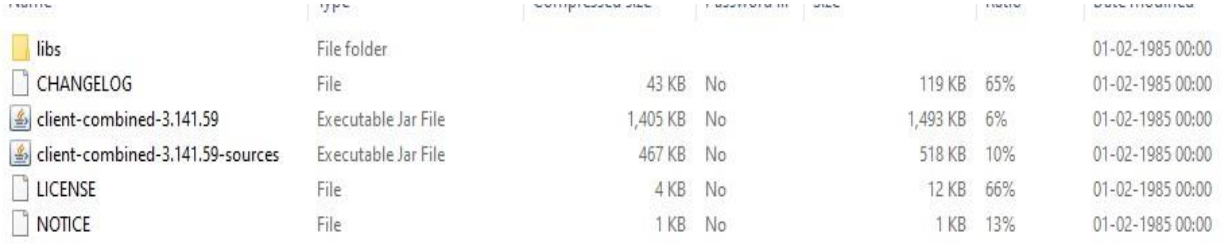
It will redirect you to a pop up window as shown below:_



3). Click (Maximize) the Highlighted Selenium option------> Select and Download the Selenium Java zip folder.



4). The downloaded file would be in zipped format. Unpack the contents in a convenient directory or folder. It contains the essential jar files required to configure Selenium WebDriver in Eclipse IDE.



libs	File folder					01-02-1985 00:00
CHANGELOG	File	43 KB	No	119 KB	65%	01-02-1985 00:00
client-combined-3.141.59	Executable Jar File	1,405 KB	No	1,493 KB	6%	01-02-1985 00:00
client-combined-3.141.59-sources	Executable Jar File	467 KB	No	518 KB	10%	01-02-1985 00:00
LICENSE	File	4 KB	No	12 KB	66%	01-02-1985 00:00
NOTICE	File	1 KB	No	1 KB	13%	01-02-1985 00:00

4. Configure Selenium WebDriver

1). Now we will configure our Eclipse IDE with Selenium WebDriver. In simple terms, we will create a new Java Project in Eclipse and load all the essential jar files in order to create Selenium Test Scripts.

2). To the earlier Created new Project, **create Package and Class**.

For That:-

PACKAGE CREATION:-

Right Click the Project------->Select **New**------>Select **Package**------>write the package name in small letters------>**FINISH**.

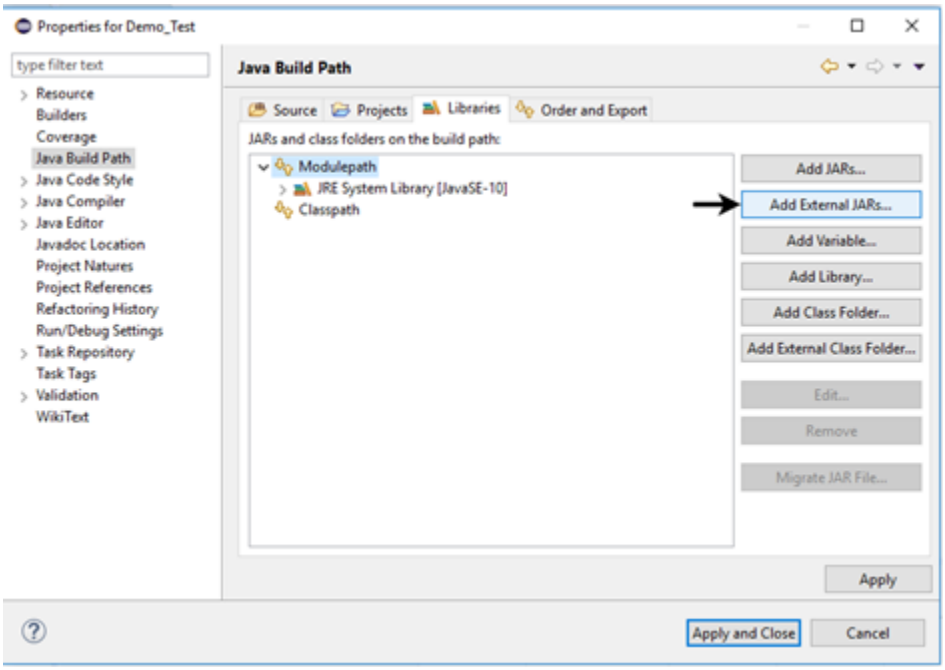
Package will be created under src folder in the project.

CLASS CREATION:-

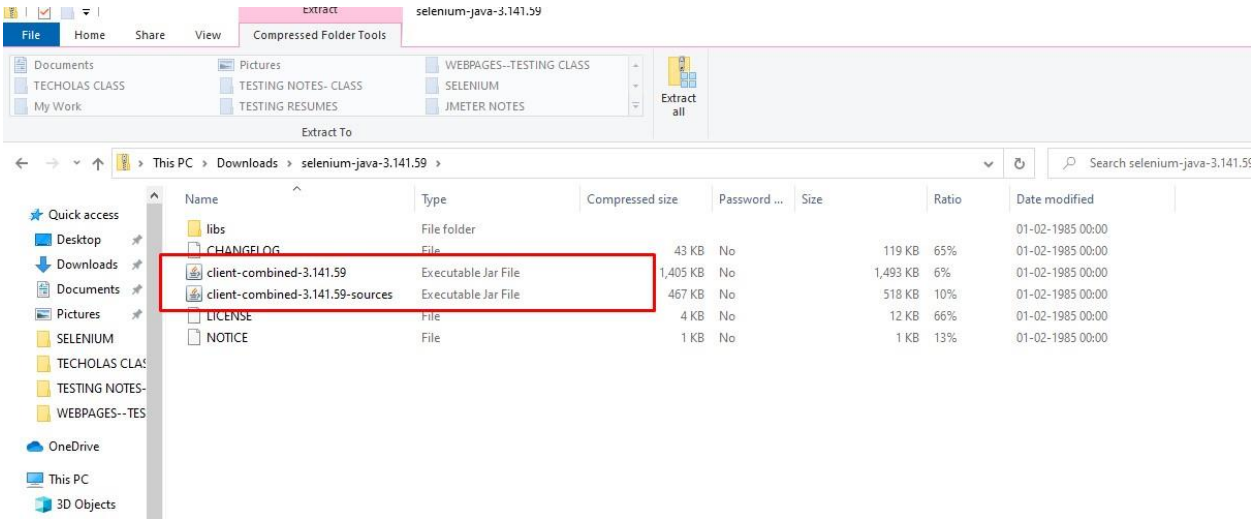
Right Click the Package------->Select **New**------>Select **Class**------>write the Class name(first letter caps on)------>Select public static void main------->**FINISH**.

3). We will **add the Selenium jar files** in our Test Suite(project).

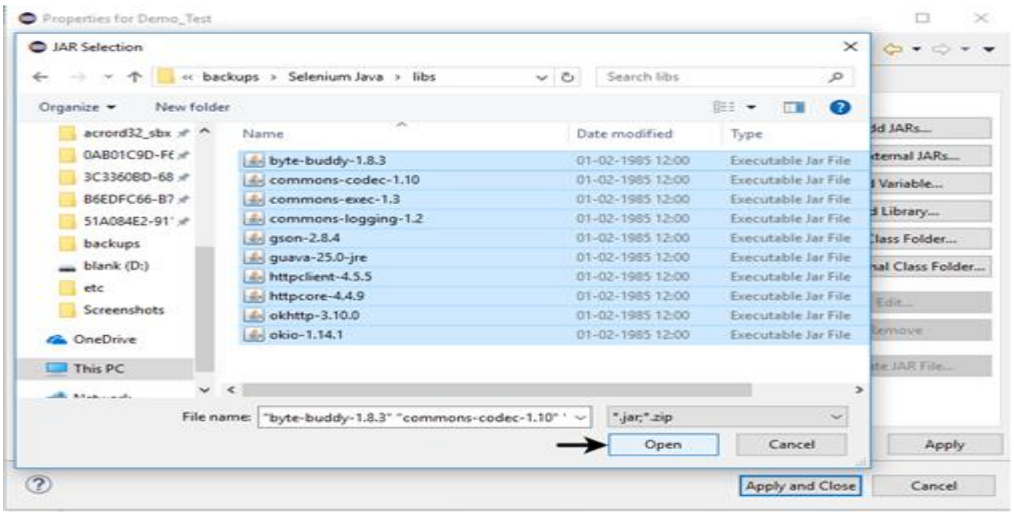
Right Click on Project-------->Select Build Path-------->Click Configure build path----->Select Libraries------>Select Classpath-------->Click Add External Jars



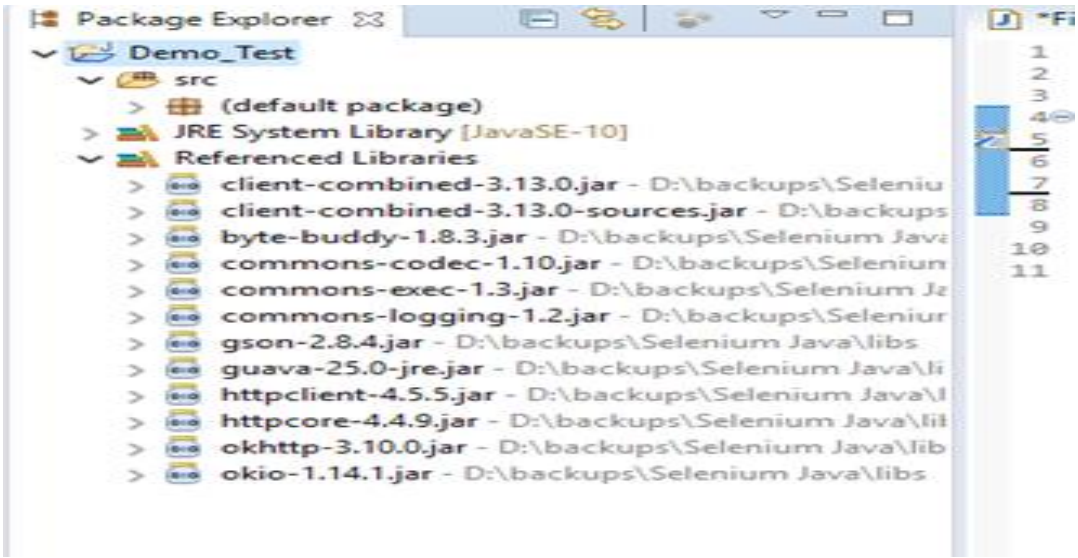
- Locate the directory where you have downloaded the Selenium jar files, select the respective jars and click on "Open" button.



- Repeat the same steps for the jars which are present under the **"libs"** folder.
- Open "libs" folder, select all of the respective jar files and click on "Open" button.



- Once you get all the Selenium jar files in your Libraries tab, click on Apply and Close button.
- The following image shows the directory structure of our "Project" test suite after adding Selenium jars.



Hence, we have successfully configured Selenium WebDriver with Eclipse IDE.

But to run our test script, we have to Download Chrome Driver for Google Chrome(if we are using Chrome as the browser). If we are using any other browser then we have to download the driver according to the browser we are using, for example for Mozilla we have to download Gecko Driver.

For that follow the following steps:-

- Check the **Version** of your Chrome Browser
- Downloadad the Chrome Driver from the following url: https://chromedriver.chromium.org/downloads
- Select the version according to your updated Chrome Browser. It will redirect you to a new webpage.
- Select the link **" chromedriver_win32.zip "**



- Download the driver, extract the zip file and save it in any location.

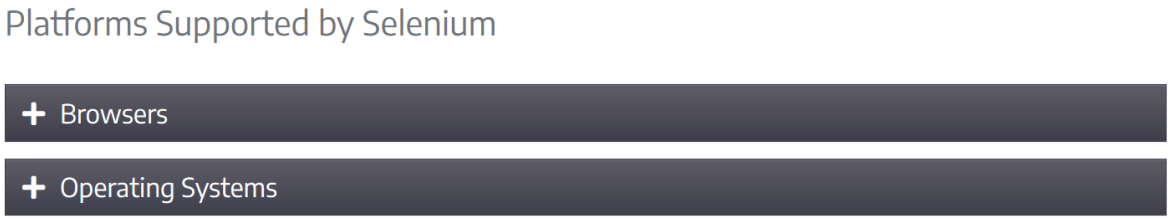
Now, we are ready to write our test scripts in Eclipse and run it in WebDriver.

Download Selenium for java In brief

Step 1. https://www.selenium.dev/downloads/

Step 2 . Scroll down and select https://github.com/SeleniumHQ/selenium/releases/download/selenium-4.4.0/selenium-java-4.4.0.zip

Step 3 . Scroll down and select Browsers



Step 4 . Expand “Browsers” - Then select Chrome documentation.
https://chromedriver.chromium.org/

Step 5. Choose Latest **stable** release ChromeDriver

Step 6 . Download the setup .https://chromedriver.storage.googleapis.com/104.0.5112.79/chromedriver_win32.zip

Step 7.Download Webdriver manager https://jar-download.com/artifact-search/webdrivermanager

Step 8 . Download ebdrivermanager from group io.github.bonigarcia (version 5.3.0)

Step 9 . Search and download “Eclipse committers for eclipse”https://www.eclipse.org/downloads/packages/

Step 10 . Download https://www.eclipse.org/downloads/download.php?file=/oomph/epp/2022-06/R/eclipse-inst-jre-mac64.dmg

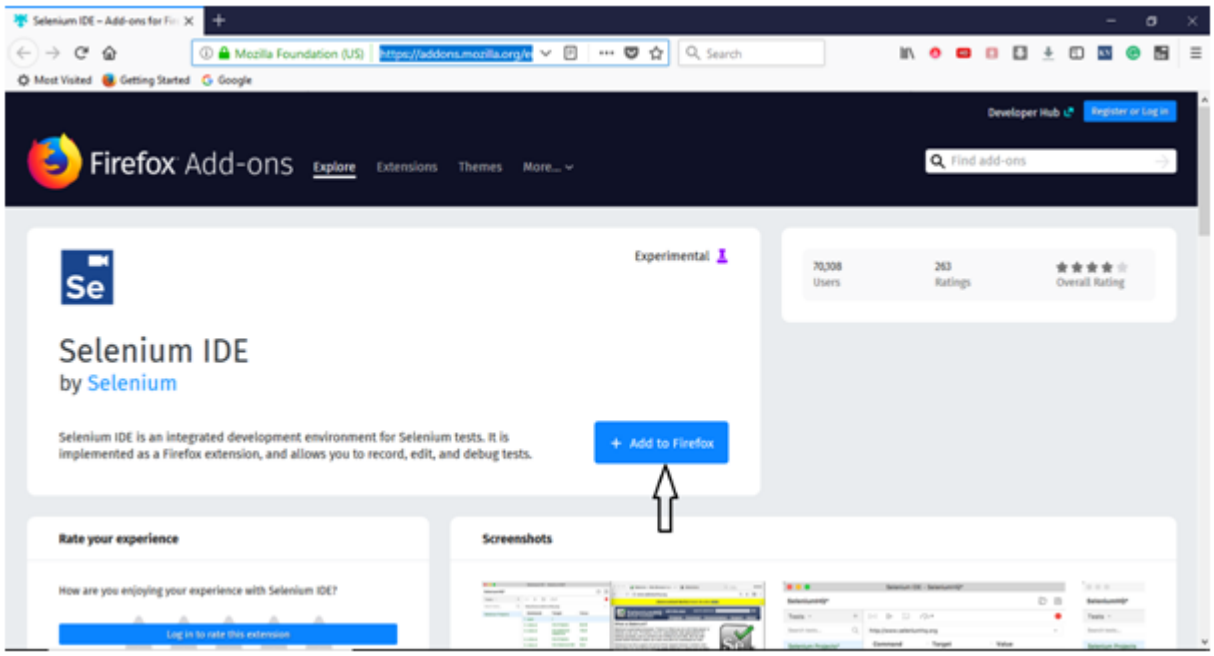
Selenium IDE-Installation

Launch Mozilla Firefox browser.

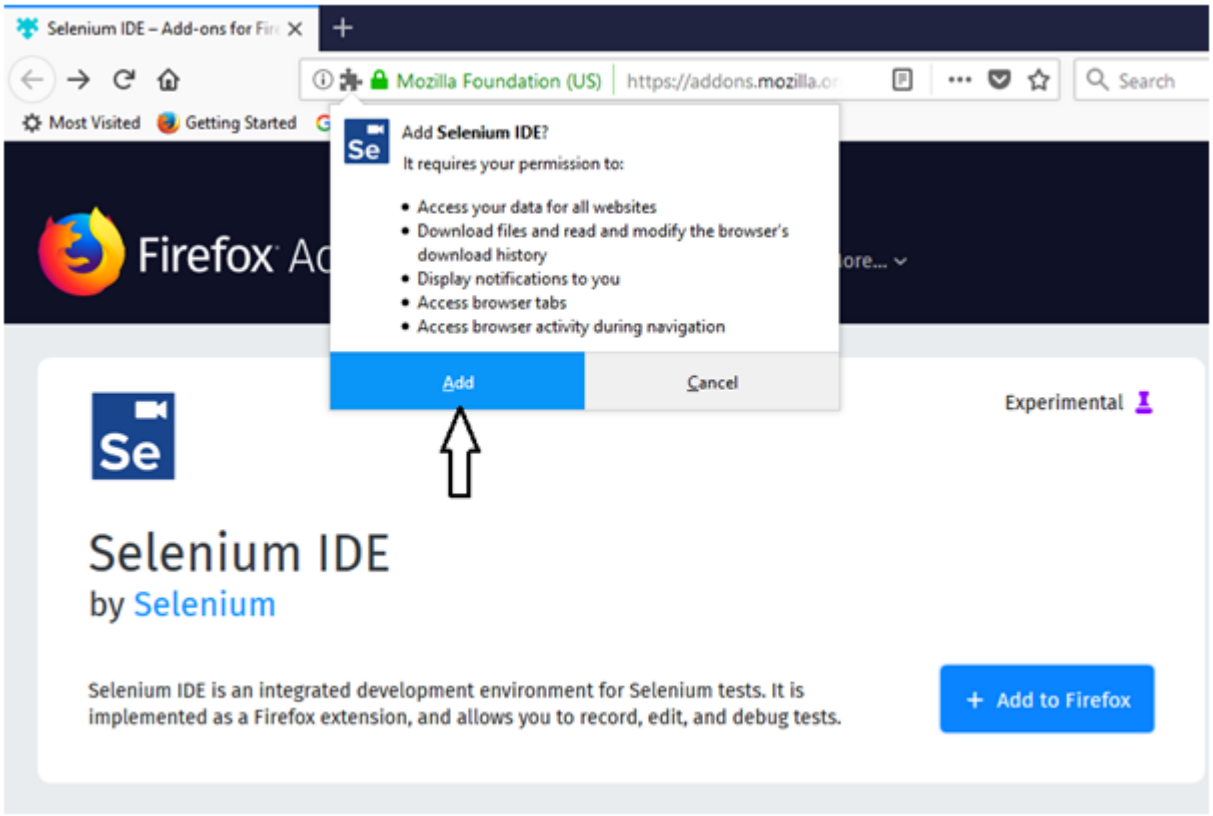
Open URLhttps://addons.mozilla.org/en-us/firefox/addon/selenium-ide/

- It will redirect you to the official add-on page of Firefox.

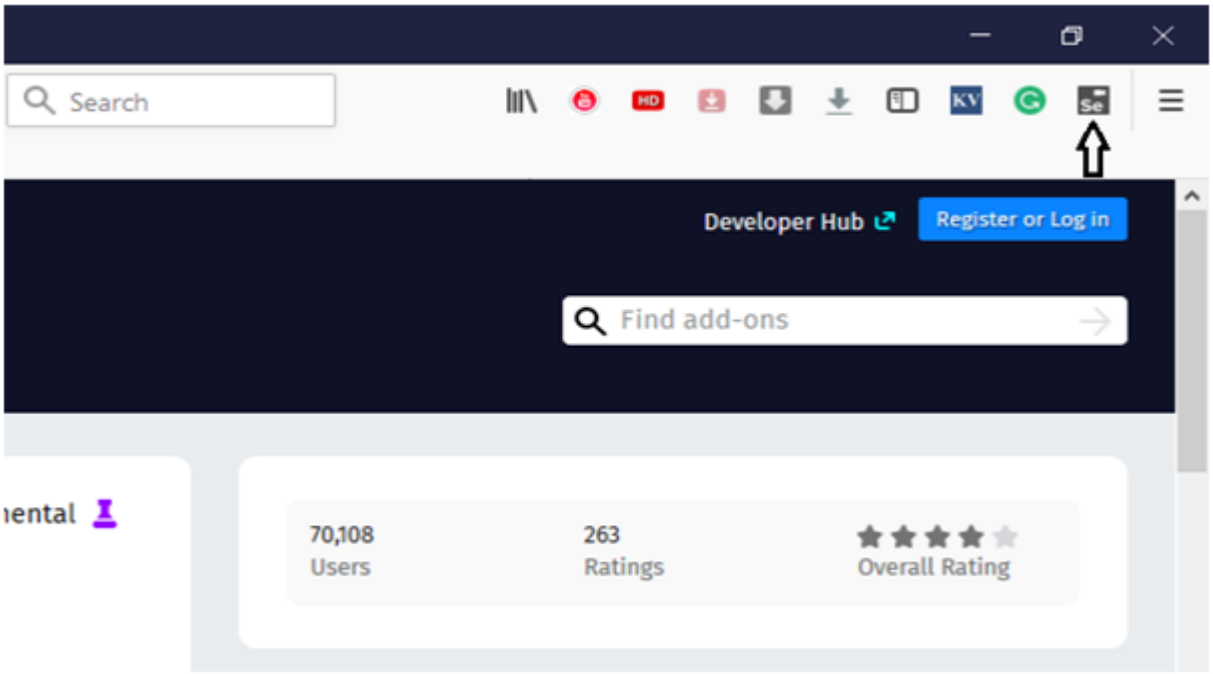
Click on "Add to Firefox" button.



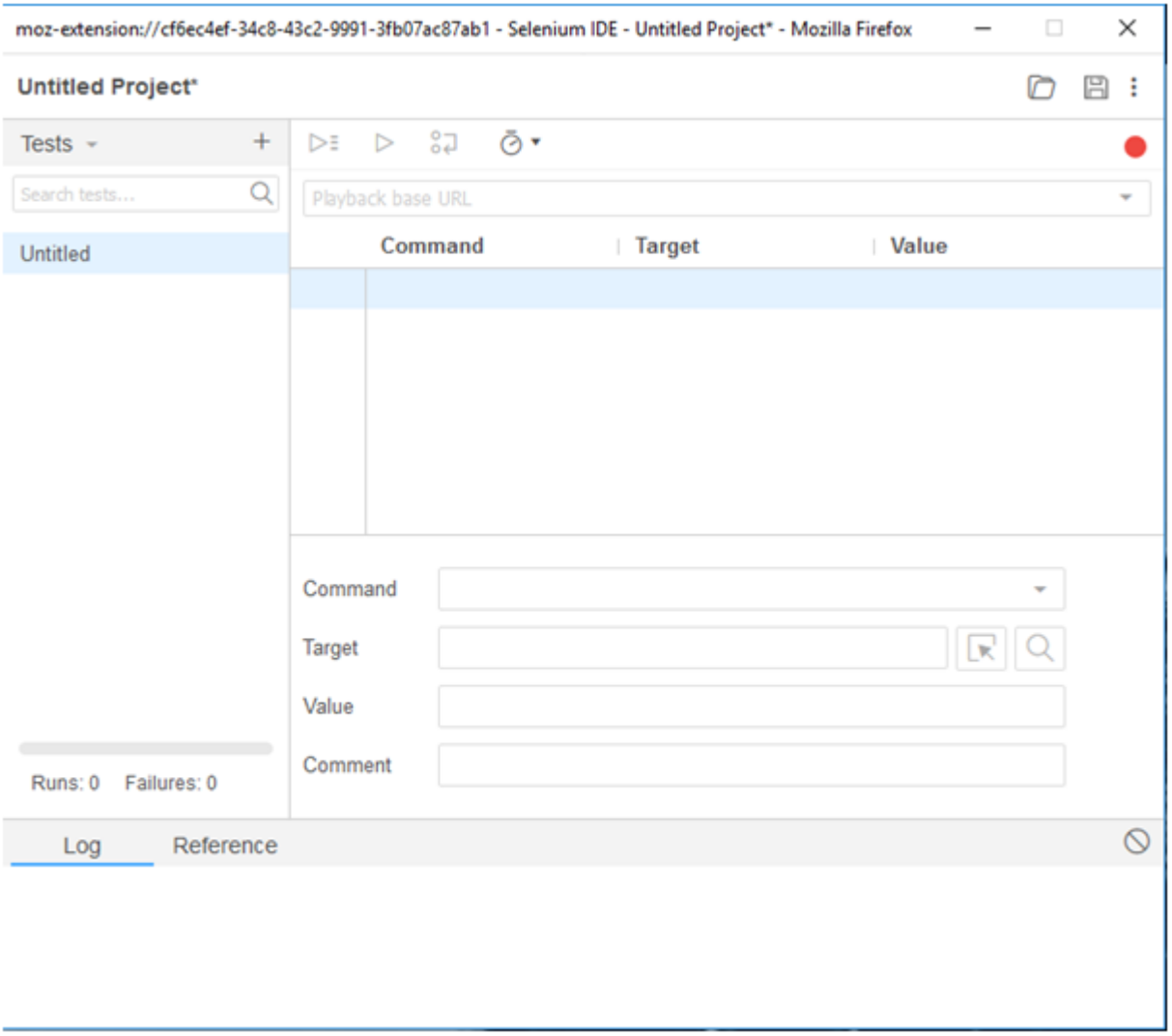
- A pop-up dialog box will appear asking you to add Selenium IDE as extension to your Firefox browser.
- Click on the "Add" button.



- Restart you Firefox browser.
- Go to the top right corner on your Firefox browser and look for the Selenium IDE icon.


Search
Developer Hub
Register or Log in
Find add-ons
70,108
Users
263
Ratings
Overall Rating

- Click on that icon to launch Selenium IDE. —-



Selenium ID - (Record and Playback tool)

Selenium IDE provides you with a GUI (Graphical User Interface) for easily recording your interactions with the website

Selenium ID Commands - Selenese

Can save as Test case and Test Suit

Test case - Single test case - saved in HTML format

Test Suit - Collection of Test cases - saved in FILE format

DEBUGGING COMMANDS

1. SET/CLEAR START POINT – (Shortcut -)STo set a point from where we have to run the
2. TOGGLE BUTTON/BREAKPOINT(Shortcut -)
3. EXECUTE THIS COMMAND (Shortcut -B)

JMETER

JMeter Installation

It is required to install 'jdk 1.6' or above on machine and successfully configured the environment variables required to run and compile java programs.

However, to verify whether you have Java installed on your system. Open your command promt and execute the following java commands as per the operating system you are working on.

Operating System	Task	Command
Windows	Open Command Console	c:\> java -version
Linux	Open Command Terminal	$ java -version
Mac	Open Terminal	machine: ~ javaTpoint$ java -version

The latest version of JMeter can be downloaded from:

https://jmeter.apache.org/download_jmeter.cgi

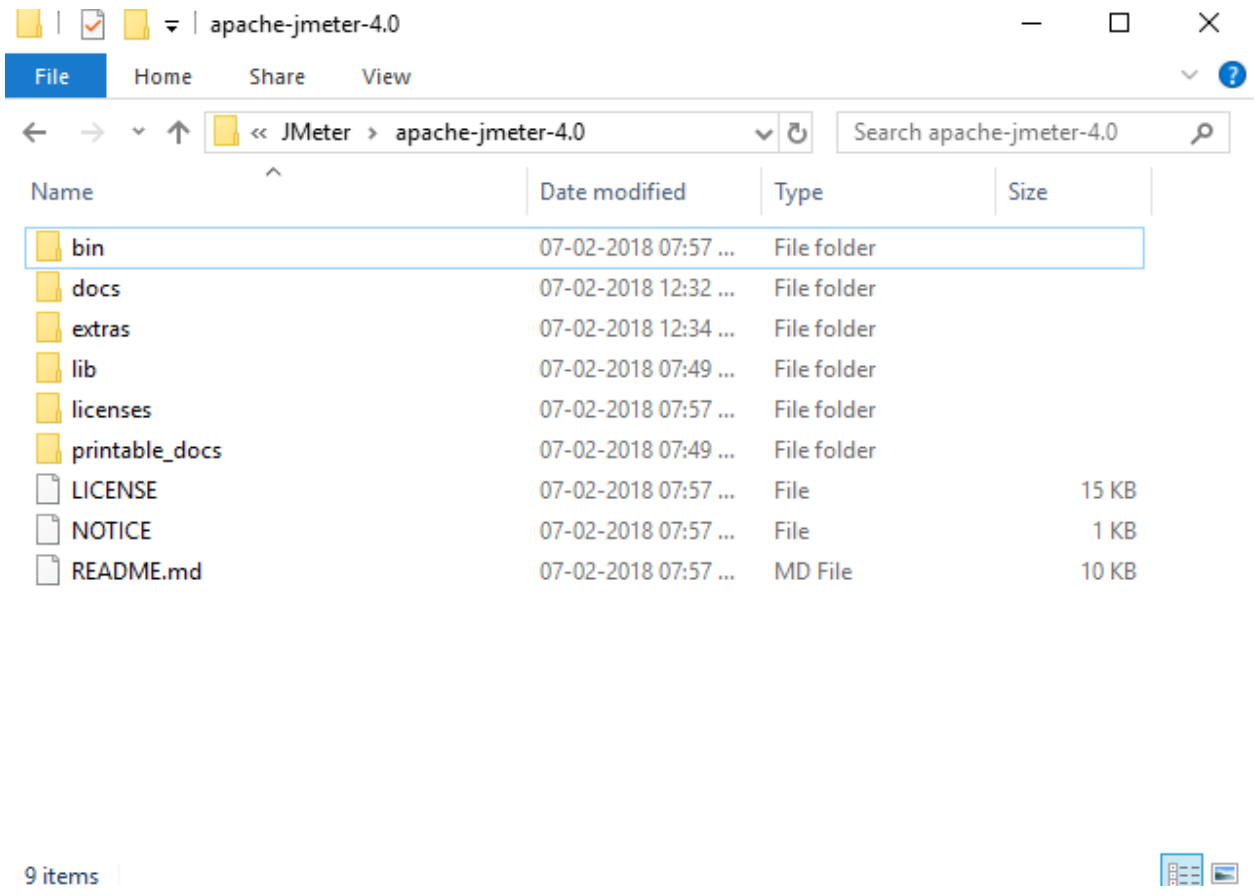
Download Latest version of JMeter

- There are two versions of JMeter available. Download according to the operating system you are working on.

File	Description
apache-jmeter-4.0.tgz	JMeter for Linux/Unix.
apache-jmeter-4.0.zip	JMeter for Windows

- After downloading JMeter, you need to extract the files at a convenient directory OR copy the extracted folder to Program Files in C-DRIVE.

- The following image shows the structure of the directories and files -



To launch JMeter in GUI mode - go to the bin directory and double-click on the respective executable file as per the operating system you are working on. For example:

Operating System	Task
Windows	Click on ApacheJMeter.jar
Linux	Click on ApacheJMeter.sh
Mac	Click on ApacheJMeter.sh

- The following image shows the default Apache JMeter interface -

